

WEEK 3



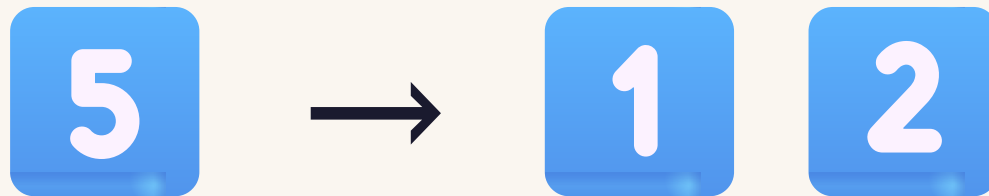
Variables

One name. Change it everywhere.

변수 — 숫자에 이름 붙이기 · 영어코딩

BIG PICTURE

Change one number, rebuild it all ✨



Set `size = 5`. Change to 12. The whole build resizes.

숫자 하나만 바꾸면 전체가 바뀝니다

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a variable?

A **name** that holds a value you can use later.

size

5

height

6

변수 = 값을 담는 이름표

WATCH IT!

The value flies into the box

size

```
range(size) # →
```

size에 5를 저장 → 어디서든 size 사용

CODE

Use the variable in a loop

```
size = 5

for i in range(size):
    agent.place(DOWN)
    agent.move(FORWARD, 1)
```

The loop runs `size` times — here, 5.

CODE

Change one number

```
# build a size-5 wall  
size = 5  
  
# now make it huge – change ONE line  
size = 12  
for i in range(size):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

No retyping the loop — just edit `size`.

PREDICT 🤖

How many blocks?

Predict first — then click reveal

```
size = 3
for i in range(size):
    agent.place(DOWN)
    agent.move(FORWARD, 1)
size = 6
for i in range(size):
    agent.place(DOWN)
    agent.move(FORWARD, 1)
```

Reveal answer 🔍 reveal

VOCAB

New words today

- variable** a name that stores a value
- value** the data the variable holds (like 5)
- =** store the right side into the name on the left
- reuse** use the same variable in many places

변수

값

대입

재사용

 DEBUG

What's wrong here?

buggy

```
# goal: build a row of `size` blocks, size is 5  
size = 5  
for i in range(3):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

Think first — then click next.



DEBUG

The fix

Bug: The loop uses the number `3`, not the variable `size`. It builds 3 blocks, not 5. Put `size` inside `range()`.

fixed

```
size = 5
for i in range(size):
    agent.place(DOWN)
    agent.move(FORWARD, 1)
```

YOUR TURN!

Resize with one variable

- ✓ Make a variable `size = 5`
- ✓ Build a wall that uses `range(size)`
- ✓ Change `size` to 12 and run again
- ✓ Compare the two walls
- ✓ Send a photo + walkthrough on KakaoTalk



QUIZ

QUESTION 1 OF 3

What is a variable?

A type of block.

A command that turns the agent.

A loop.

A name that stores a value you can reuse.

 QUIZ

QUESTION 2 OF 3

What's wrong?

```
size = 5
for i in range(3):
    ...
```

size can't be 5.

range needs two numbers.

Nothing is wrong.

The loop uses 3, not size — it builds 3 blocks.



QUIZ

QUESTION 3 OF 3

Why use a variable for the size?

It changes the block color.

It makes the agent faster.

Change one number and the whole build resizes.

Variables are required by Minecraft.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Variables down!

Next: nested loops — a loop inside a loop.

다음 주: 반복문 안의 반복문